

NAVLEEN SAINI

FULL STACK DEVELOPER

431-277-9876 | navanmol777@gmail.com | GitHub | Linked In

7 Fossil Creek Crescent, Winnipeg, MB | R3Y 2K1

PORFESSIONAL SUMMARY

Applied Computer Science student with strong foundation in algorithms, data structures, and backend development. Experienced in building server-side applications using Node.js and Express, implementing custom data structures in C++, and designing relational databases using SQL. Interested in software engineering and intelligent systems.

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript

Backend: Node.js, Express.js

Core CS: Data Structures, Algorithms, OOP, Time & Space Complexity Analysis

AI Concepts: Search Algorithms (BFS, DFS, A*, heuristic search)

Database: SQL, Relational Database Design

Tools & Design: Git, Figma

Other: Physical Computing, Hardware-Software Integration

PROJECTS

Backend-Focused Web Application (Node.js, Express, MongoDB, JavaScript)

- Implemented user authentication and authorization using secure password hashing (bcrypt, salting) and protected routes.
- Developed custom middleware for request validation and access control.
- Integrated MongoDB using Mongoose for schema modeling and database interaction.
- Applied input validation and server-side error handling to ensure data integrity and security.
- Structured project using layered architecture (routes, controllers, models) for maintainability.

Autonomous Safety Robot (Arduino, C++)

- Programmed robot with real-time control and obstacle avoidance.
- Applied optimized C++ functions and signal smoothing for faster, smoother operation.

Fast search AI research (on going | python)

- Implementing heuristic search algorithms including A* and binary search in Python.
- Designing optimized solutions for interval scheduling and interval partitioning problems.
- Analyzing algorithm efficiency and exploring integration of AI search methods into backend systems.

Simple Portfolio deployment (html, CSS)

Written code basic html and css for portfolio and deployed using GitHub.

EDUCATION AND CERTIFICATION

B.sc in Applied Computer Science | University of Winnipeg (2023 - 2026)

Major GPA: 3.7 | CGPA: 3.4

EXPERIENCE

Overnight manger | Tim Hortons (2023 - 2026)

Teaching Assistant | NYC ACS (2022 - 2023)